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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2024

IS1101-1- FUNDAMENTALS OF CYBER SECURITY
(For ACT, EC, EE, VLSI, AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the definition of cybercrime?**
A) Any illegal activity involving the use of physical force
B) Any illegal behavior directed by means of electronic operations that targets the security of computer systems and the data processed by them
C) Legal activities conducted online
D) Cyber activities conducted for entertainment purposes
2. **What is a "Salami Attack"?**
A) A widespread attack using viruses
B) Attacks on food supply systems
C) Minor and unnoticed changes in financial transactions to commit financial crimes
D) Large-scale denial-of-service attacks
3. **Who are typically classified as "Type III" cybercriminals?**
A) Hackers seeking recognition
B) State-sponsored hackers
C) Terrorist organizations
D) Insiders such as disgruntled employees
4. **Which phase involves gathering information about a target without their knowledge?**
A) Passive attack
B) Active attack
C) Scanning
D) Exploitation
5. **What is the main goal of port scanning in cybercrime?**
A) To encrypt data for ransom
B) To identify open or closed ports on a target system
C) To steal sensitive information
D) To launch a denial-of-service attack
6. **What is "dumpster diving" in the context of cybercrime?**
A) Searching through trash to find confidential information
B) Using a fake website to collect personal data
C) Intercepting communication between two parties
D) Cracking passwords by trying all possible combinations
7. **Which of the following attacks is characterized by sending unsolicited messages to mobile devices?**
A) Phishing
B) Mishing
C) Smishing
D) Vishing
8. **What is the primary purpose of the IMEI number on a mobile device?**
A) To store contact information
B) To encrypt data stored on the device
C) To track the phone's location.
D) To identify the device on a GSM network

9. The first computer virus, named as _____, was a stealth virus
 - A) Heart
 - B) Lungs
 - C) Brain
 - D) WannaCry
10. Man-in-the middle attack is an _____ attack in passwords cracking
 - A) Brakeforce attack
 - B) Online attacks
 - C) Brute force attack
 - D) Offline attacks
11. _____ infects the storage media on which OS is stored and which is used to start the computer system
 - A) Program viruses
 - B) Stealth viruses
 - C) Boot sector viruses
 - D) Macroviruses
12. DoS attack primarily uses _____ to attack a system
 - A) RAM overflow technique
 - B) CPU overflow technique
 - C) Hard disk overflow technique
 - D) Cache Overflow technique
13. _____ phishers do not identify specific prospective victims in advance
 - A) Rod-and-reel
 - B) Dragnet
 - C) Lobsterpot
 - D) Gillnet
14. Which one of the following is not a purpose of password cracking
 - A) As a preventive measure by system administrators to check for easily crackable passwords.
 - B) To gain unauthorized access to a system
 - C) To erase password
 - D) To recover a forgotten password
15. Last step in Phishing is
 - A) Setup
 - B) Collection
 - C) Attack
 - D) Identity theft and fraud
16. Ping sweep is involved in which stage of an attack
 - A) Network probe
 - B) Crossing the line toward electronic crime
 - C) Grab the data
 - D) Capturing the network
17. In digital forensics, what does the "chain of custody" concept refer to?
 - A) The process of securing and documenting evidence to maintain its integrity
 - B) The sequence of actions taken during a forensic analysis
 - C) The chronological documentation trail, etc. that indicates the seizure, custody, control, transfer, analysis and disposition of evidence (physical or electronic)
 - D) The hierarchy of law enforcement agencies involved in a cybercrime investigation
18. What is the phase of the digital forensics' life cycle, where the findings and conclusions are documented for presentation in court?
 - A) Analysis
 - B) Examination
 - C) Reporting
 - D) Collection
19. What is the primary objective of digital forensics?
 - A) To recover lost data from computer systems
 - B) To create software for analyzing digital data
 - C) To prevent cyberattacks
 - D) To investigate and analyze digital evidence for legal purposes
20. In email forensic analysis, what is the significance of examining email headers?
 - A) To determine the sender's emotional state
 - B) To check for grammar and spelling errors
 - C) To identify the email recipient's location
 - D) To trace the email's origin and route

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define cybercrime. Explain the two types of cybercrimes with an example for each.	8	L*2	1	1
	b) Explain Bluejacking and Bluesnarfing. List the 5 security strategies a company can implement to secure their mobile devices.	8	L2	2	1

IS1101-1**SEE – December 2024**

2.	a)	Define child pornography. Explain how pedophiles operate.	8	L2	1	1
	b)	Explain the following cybercrimes/cyberattack briefly.				
		i) Email spoofing				
		ii) Spamming				
		iii) Password sniffing				
		iv) Computer sabotage	8	L2	2	1
3.	a)	Explain the terms Logic bomb, E-Mail bombing, industrial espionage and Salami attack.	8	L2	1	1
	b)	Discuss the types and techniques of Credit Card Frauds.	8	L2	2	1

Unit – II

4.	a)	Discuss deceptive phishing, malware-based phishing, keyloggers and session hijacking.	8	L2	4	1
	b)	What is Backdoor? What does a Backdoor do?	8	L2	3	1
5.	a)	Write a note on each of the following				
		i. Boot Sector viruses				
		ii. Program Viruses				
		iii. Stealth Viruses				
		iv. Polymorphic viruses	8	L2	3	1
	b)	Discuss the classification of DoS attacks.	8	L2	4	1
6.	a)	List and explain four types of identity theft, along with four effective countermeasures to prevent it.	8	L2	4	1
	b)	Explain Dragnet and Lobsterpot phishing methods. Discuss any four Phishing Countermeasures.	8	L2	3	1

Unit – III

7.	a)	List 8 guidelines for the digital evidence collection phase.	8	L2	5	1
	b)	List forensics life cycle phases. Explain the first two phases.	8	L2	5	1
8.	a)	What is chain of custody? Why is it used? Discuss the documents that are included in chain of custody.	8	L2	5	1
	b)	Explain the technical challenges involved Computer Forensics by considering Raw data and its structure.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
